DAY 35 | HANDS-ON ACTIVITIES

Drift Detection and Optimization: Problem Statements

Seven problems using simulated reference and current periods with planted drift.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Name the Drift

Level: Warm-up | **Concepts:** drift types

Classify: more Japanese questions; refund period changes; index not rebuilt for two weeks; provider updates the model; answers getting longer; a new product launches.

Sample run / expected result

```
More Japanese: data drift
Refund change: concept drift
Stale index: knowledge drift
Provider update: model drift
```

Hints

- Ask: what changed, the input, the truth, the documents or the model?

Problem 2: PSI on Question Length

Level: Easy | **Concepts:** PSI

Compute PSI for question length between week 1 and week 4, and between two halves of week 1 as a sanity check.

Sample run / expected result

```
week1 vs week4: PSI 0.39 (significant)
week1 half A vs B: PSI 0.00 (stable)
```

Hints

- Always sanity-check a drift metric on two samples that should match.

Problem 3: Categorical Drift

Level: Easy | **Concepts:** PSI for categories

Compute PSI for topic and for language share between the two periods. Which contributes most?

Sample run / expected result

```
topic PSI 1.28 (visa appeared)
language PSI 0.22 (more Japanese)
```

Hints

- Use the same category list for both periods.

Problem 4: Embedding Drift

Level: Medium | **Concepts:** centroids, clustering

Compute the centroid drift of question embeddings per week and plot it. Find the week the drift jumps.

Sample run / expected result

```
week 2: 0.03 | week 3: 0.01 | week 4: 0.37 (jump)
```

Hints

- Compare with a same-period baseline to know what normal looks like.

Problem 5: Concept Drift: Update the Golden Set

Level: Medium | **Concepts:** golden set maintenance

The refund period changed from 30 to 14 days. Show the old golden item now passes for a wrong answer, update it, and confirm the old answer fails.

Sample run / expected result

```
old golden: "30 days" answer PASSES (wrong)
updated golden: "30 days" answer FAILS (correct)
```

Hints

- Golden sets drift too.

Problem 6: Optimisation Experiments

Level: Challenging | **Concepts:** one change at a time

On the golden set with new visa questions, compare: baseline; k=5; adding the missing visa document; both. Report hit rate per experiment.

Sample run / expected result

```
baseline 0.80 | k=5 0.80 | add visa doc 1.00 | both 1.00
```

Hints

- The data fix usually beats tuning.

Problem 7: Weekly LLMOps Job

Level: Stretch | **Concepts:** automation

Write weekly_job.py that computes drift metrics and golden set results, writes a Markdown report and exits with code 1 if any threshold is breached.

Sample run / expected result

```
report_2026-W52.md written
psi_topic 1.28 > 0.25 -> exit 1 (review needed)
```

Hints

- Schedule it with an Azure DevOps scheduled pipeline or a cron job.

Henka (変化): change